

0. ABSTRACT ALGEBRA PREREQUISITES

NOTATION

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ (integers)
- $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$
- $\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}$ (e.g. $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$)
- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ (non-negative integers)
- $\mathbb{P} = \{1, 2, 3, \dots\}$ (positive integers)
- $\mathbb{Q} = \{\text{rational numbers}\}$
- $\mathbb{R} = \{\text{real numbers}\}$
- $\mathbb{C} = \{\text{complex numbers}\}$
- $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^* = \mathbb{Q} - \{0\}, \mathbb{R} - \{0\}, \mathbb{C} - \{0\}$
- $GL_n(\mathbb{R}) = \{n \times n \text{ invertible matrices w/ entries in } \mathbb{R}\}$ ("general linear group")
- $SL_n(\mathbb{R}) = \{M \in GL_n(\mathbb{R}) \mid \det(M) = 1\}$ ("special linear group")

GROUP

DEFINITION

A group $(G, *)$ is a set G with an operation $*$, s.t. the following hold:

- Closure: $a * b \in G \quad \forall a, b \in G$
- Associativity: $a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$
- Identity: $\exists e \in G$, s.t. $a = a * e = e * a \quad \forall a \in G$
- Inverse: $\forall a \in G, \exists b \in G$, s.t. $a * b = b * a = e$

Note: Commutativity is not on there

EXAMPLES OF GROUPS

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ with $+$
- $\mathbb{Q}^*, \mathbb{R}^*, \mathbb{C}^*$ with $\times$
- $5\mathbb{Z} = \{5a \mid a \in \mathbb{Z}\}$ with $+$
- $\{1, -1\}$ with $\times$
- $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ with $+$ mod 6
- $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$ with $\times$ mod 7

EXAMPLES OF NOT GROUPS

- $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$ with $+$ $\leadsto$ not closed
- $\mathbb{P}$ with $+$ $\leadsto$ no identity
- $\mathbb{Z}$ with $-$ $\leadsto$ not associative
- $2\mathbb{Z} + 1 = \{2a + 1 \mid a \in \mathbb{Z}\}$ with $\times$ $\leadsto$ no inverse

PROPOSITION

- The identity e of G is unique

Proof: If e_1, e_2 are identities, then $e_1 = e_1 * e_2 = e_2$ e_2 : identity e_1 : identity

- The inverse b of each $a \in G$ is unique $\leadsto$ denoted as a^{-1}

Proof: If b, c are inverses of a , then $b = b * e = b * (a * c) = (b * a) * c = e * c = c$

- $(a^{-1})^{-1} = a \quad \forall a \in G$

- $(a * b)^{-1} = b^{-1} * a^{-1} \quad \forall a, b \in G$

- $\forall a, b, c \in G$,

$\hookrightarrow a * x = b$ and $y * a = b$ have unique solutions in G

$\hookrightarrow a * b = a * c \Rightarrow b = c$ and $a * b = c * b \Rightarrow a = c$

PERMUTATIONS

Shun/18:5 (@shun4midx)

DEFINITION

A **permutation** is a bijective function $f: T \rightarrow T$ on a set T into itself

EXAMPLE

$T = \{1, 2, 3\}$ with $f(1)=2, f(2)=3, f(3)=1$ is a permutation represented by the array $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

REMARK

Every **block cipher** with a **specific key** is a permutation

- DES_k: $\{0, 1\}^{64} \rightarrow \{0, 1\}^{64}$ (block size = 64 bits)
- AES_k: $\{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ (block size = 128 bits)

SYMMETRIC GROUP

DEFINITION

A **symmetric group** S_n is a set of all permutations on $\{1, 2, \dots, n\}$ with $\circ$ (composition) $(f \circ g)(x) = f(g(x))$

EXAMPLE

In S_3 , we have $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$

Note: $S_3 = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\}$

We may also say $S_3 = \{e, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$ (just write $(x\ f(x)\ f(f(x)) \dots)$ until $f \dots f(x) = x$)

↳ E.g. $(1\ 2\ 3)^{-1} = (3\ 2\ 1) = (1\ 3\ 2)$

ABELIAN GROUP

DEFINITION

A group $(G, *)$ is **commutative** or **abelian** if $a * b = b * a \ \forall a, b \in G$

EXAMPLES OF NON-ABELIAN GROUPS

For $n \geq 3$, S_n is NOT abelian; with matrix multiplication, $GL_n(\mathbb{R}), SL_n(\mathbb{R})$ aren't abelian

CYCLIC GROUP

DEFINITION

A group $(G, *)$ is **cyclic** if there exists a generator $g \in G$ such that every $a \in G$ is of the form $a = \underbrace{g * \dots * g}_{n \text{ copies}} = g^n$ for some $n \in \mathbb{Z}$

EXAMPLES OF CYCLIC GROUPS

- $(\mathbb{Z}, +)$ is cyclic with generators $+1$ and -1
- $(\mathbb{Z}_n^*, \otimes)$ is cyclic: $\{1 = 3^0 = 3^6, 2 = 3^2, 3 = 3^3, 4 = 3^4, 5 = 3^5, 6 = 3^3\}$
- $(\mathbb{Z}_9^*, \otimes)$ is cyclic with generators 2 and 5

EXAMPLES OF NON-CYCLIC GROUPS

- $(\mathbb{Q}, +)$
- $(\mathbb{Z}_8^*, \otimes) \cong \text{Klein } 4$

$\otimes$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

GROUP ORDER

Shun/18:5 (@shun4midx)

DEFINITION

The **order** of a group $(G, *)$, denoted as $|G|$, is the number of elements in G , or the cardinality of G

EXAMPLES

- $|S_3| = 6$, $|S_4| = 24$, $|S_n| = n!$
- $|\mathbb{Z}_p| = p$, for any prime p , $|\mathbb{Z}_p^*| = p-1$
- $|\mathbb{Z}_n^*| = |\{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}| = \phi(n) \leadsto$ Euler ϕ -function

DEFINITION

A group $(G, *)$ is finite if $|G| < \infty$, and is infinite if $|G| = \infty$

EXAMPLE

- $|\mathbb{Z}| = \infty$, $|\mathbb{Q}| = \infty$, $|\mathbb{R}| = \infty$, $|\mathbb{C}| = \infty$
countable ($|\mathbb{Z}| = |\mathbb{Q}|$) uncountable

EXAMPLE

$|GL_8(\mathbb{Z}_2)| = ?$ (i.e. 8×8 invertible matrices)

#options for the 1st col = All except only 0 = $2^8 - 1$

#options for the 2nd col = All except multiples of first col = $2^8 - 2$

#options for the 3rd col = All except $(0 \sim 7) \times 1^{\text{st}} \text{ col} + (0 \sim 7) \times 2^{\text{nd}} \text{ col} = 2^8 - 2^2$

$\vdots$

$$\therefore |GL_8(\mathbb{Z}_2)| = (2^8 - 1)(2^8 - 2)(2^8 - 2^2)(2^8 - 2^3) \dots (2^8 - 2^7)$$

SUBGROUP

DEFINITION

A subset H of a group $(G, *)$ is a **subgroup** of G if H is itself a group under the operation $*$ in G

PROPOSITION

A subset H of a group $(G, *)$ is a subgroup of G iff

- $a * b \in H \quad \forall a, b \in H$
- $a^{-1} \in H \quad \forall a \in H$

COSET

DEFINITION

If H is a subgroup of a group G , and $g \in G$, then:

- $gH := \{gh \mid h \in H\}$ is a **left coset** of H in G
- $Hg := \{hg \mid h \in H\}$ is a **right coset** of H in G

(For abelian groups with $+$, the notation is changed to $g+H$ or $H+g$)

REMARK

In general, left coset $\neq$ right coset

E.g. $(123)\{e, (12)\} = \{(123), (13)\}$; $\{e, (12)\}(123) = \{(123), (23)\}$

PROPOSITION

$$g_1H = g_2H \Leftrightarrow g_1^{-1}g_2 \in H$$

LAGRANGE THEOREM

THEOREM (LAGRANGE)

If H is a subgroup of a finite group G , then $|H|$ divides $|G|$

DEFINITION

If H is a subgroup of a finite group G , then $[G:H] := \frac{|G|}{|H|}$ is the index of H in G

DEFINITION

For $g \in G$, the order of g , denoted by $o(g)$ is the smallest positive integer n such that $g^n = e$

COROLLARY

$o(g)$ divides $|G|$

$\hookrightarrow \langle g \rangle = \{g, g^2, \dots, g^{o(g)} = e\}$ is a subgroup of G

EXAMPLE

$$|SL_2(\mathbb{Z}_7)| = \frac{1}{6} |GL_2(\mathbb{Z}_7)|$$

Proof

$$\text{Let } H = SL_2(\mathbb{Z}_7) = \left\{ \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \mid a_{ij} \in \mathbb{Z}_7, a_{11}a_{22} - a_{12}a_{21} = 1 \right\}$$

For $t=2,3,4,5,6$, define the left coset $H_t = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} H$

Then, $H_t = \{M \in GL_2(\mathbb{Z}_7) \mid \det(M) = t\}$; H and H_t 's are mutually disjoint

$f_t: H \rightarrow H_t$ defined by $f_t(M) = \begin{bmatrix} 1 & 0 \\ 0 & t \end{bmatrix} M$ is bijective (multiply by the inverse $\begin{bmatrix} 1 & 0 \\ 0 & t^{-1} \end{bmatrix}$)
 $\therefore |H| = |H_t|$

$$GL_2(\mathbb{Z}_7) = H \cup H_2 \cup H_3 \cup \dots \cup H_6 \Rightarrow |GL_2(\mathbb{Z}_7)| = 6|H| = 6|SL_2(\mathbb{Z}_7)| \quad \square$$

HOMOMORPHISM AND ISOMORPHISM

DEFINITION

$(G, *)$ and $(H, \cdot)$ are two groups

- A function $f: G \rightarrow H$ is a homomorphism if $f(a+b) = f(a) \cdot f(b) \quad \forall a, b \in G$
- A function $f: G \rightarrow H$ is an isomorphism if f is a bijective homomorphism. We say G is isomorphic to H , or $G \cong H$.
 $\hookrightarrow$ 1-1 + onto

EXAMPLES OF HOMOMORPHISM

- $f(\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ with $f(1)=2 \Rightarrow$ injective, not surjective
- $f(\mathbb{Z}, +) \rightarrow (\mathbb{Z}_7^*, \times \bmod 7)$ with $f(1)=3 \Rightarrow$ surjective, not injective

EXAMPLES OF ISOMORPHISM

- $f: (\mathbb{C}^*, \times) \rightarrow (\mathbb{C}^*, \times)$ defined by $f(atbi) = a-bi$
- $f: (\mathbb{Z}_6, + \bmod 6) \rightarrow (\mathbb{Z}_7^*, \times \bmod 7)$ with $f(1)=3$